

Name : Rutuja Gajanan Waghamare

USN : 2VX24UE043

Expt : 01

Title : Selection Sort

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In [1]: import random
import time
import matplotlib.pyplot as plt

def selection_sort(arr):
    n = len(arr)
    for i in range(n):
        min_idx = i
        for j in range(i+1, n):
            if arr[j] < arr[min_idx]:
                min_idx = j
        arr[i], arr[min_idx] = arr[min_idx], arr[i]

    return arr

arr = list(map(int, input("Enter the list of elements separated by space: ").split()))
print("Input array: ", arr)

sorted_arr = selection_sort(arr)
print("Sorted array:", sorted_arr)

start_time = time.time()
sorted_arr = selection_sort(arr)
end_time = time.time()
print("Time taken to sort:", end_time - start_time, "seconds")
print("=====")

n_values = [5000, 6000, 7000, 8000]
time_values = []

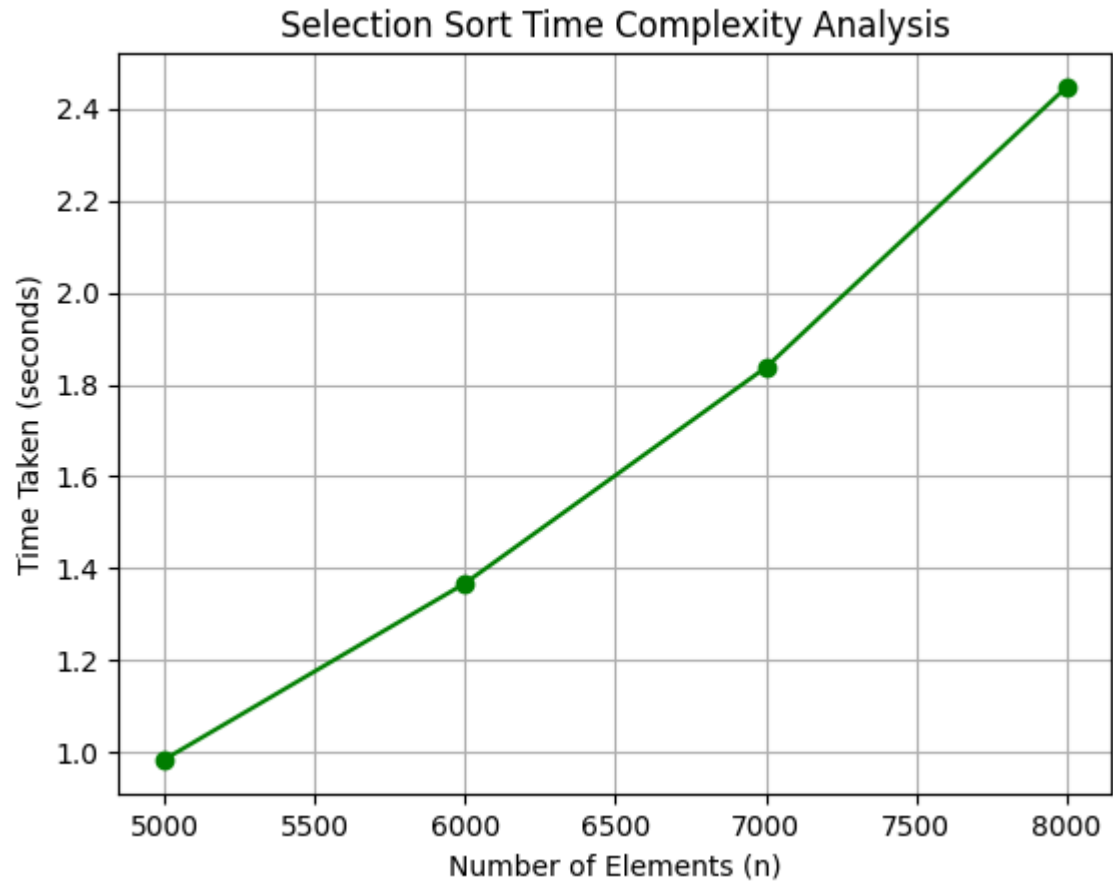
for n in n_values:
    arr = [random.randint(1, 9) for _ in range(n)]

    start_time = time.time()
    sorted_arr = selection_sort(arr)
    end_time = time.time()

    time_taken = end_time - start_time
    print("Time taken to sort", n, "elements:", time_taken, "seconds")
    time_values.append(time_taken)
print("=====")

plt.plot(n_values, time_values, color = 'green', marker='o')
plt.xlabel('Number of Elements (n)')
plt.ylabel('Time Taken (seconds)')
plt.title('Selection Sort Time Complexity Analysis')
plt.grid(True)
plt.show()
```

```
Input array: [14, 7, 8, 26]
Sorted array: [7, 8, 14, 26]
Time taken to sort: 0.00013184547424316406 seconds
=====
Time taken to sort 5000 elements: 0.9836430549621582 seconds
Time taken to sort 6000 elements: 1.3672618865966797 seconds
Time taken to sort 7000 elements: 1.8368117809295654 seconds
Time taken to sort 8000 elements: 2.4472014904022217 seconds
=====
```



In []: